

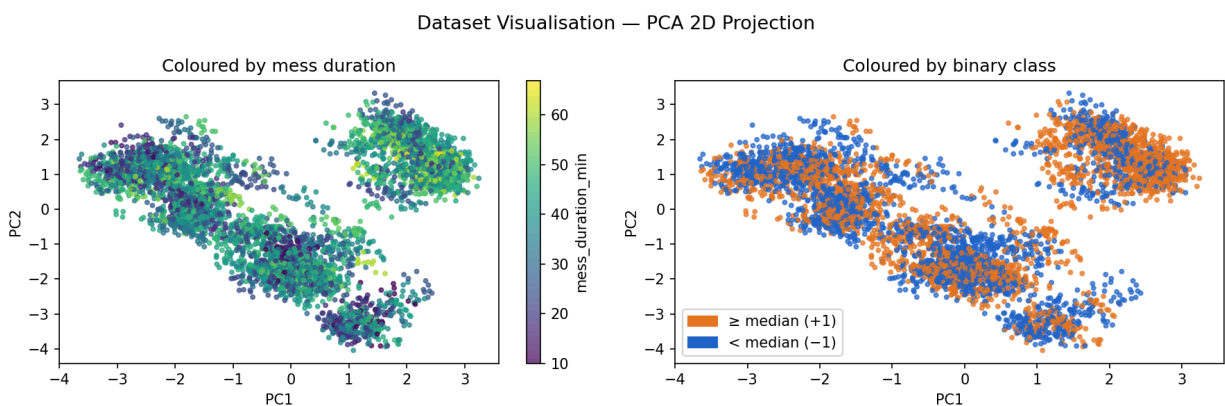
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Question - 5

Experiments

Using the dataset you have collected at Question 4

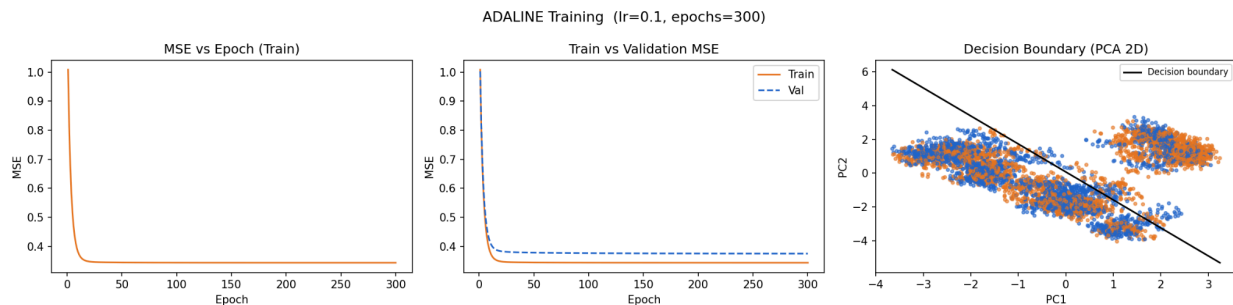
A. Plot the dataset



Observation - The PCA projection shows that the data naturally clusters into a few groups. Looking at the continuous values, there's a clear gradient where shorter durations sit on the left (negative PC1) and longer durations move toward the right (positive PC1). When we simplify this into two categories, the groups start to separate, but they still overlap quite a bit in the middle. This "messy" middle ground proves that the classes aren't perfectly distinct, which is exactly why we need a well-tuned model like ADALINE. Because ADALINE works to minimize the Mean Squared Error, it's much better at finding the most accurate boundary through that crowded center than a basic classifier would be.

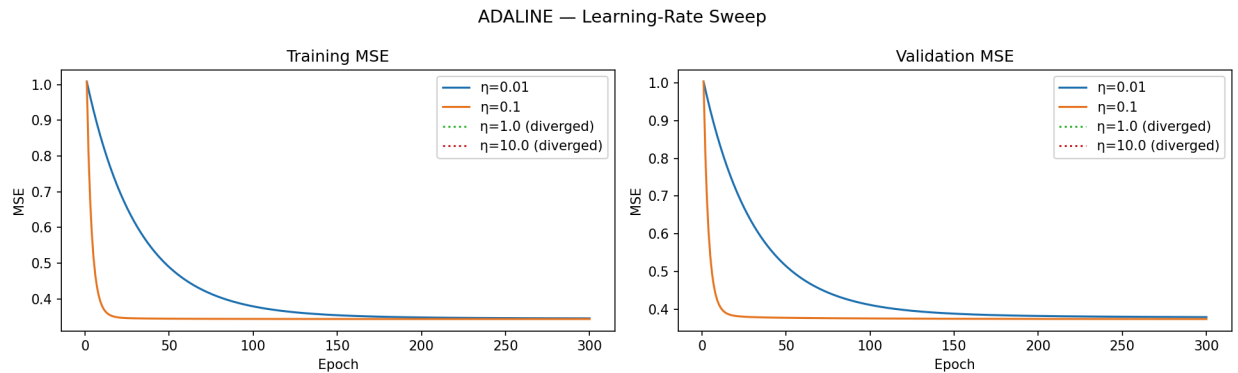
B. Train Adaline and plot:

- Mean-Squared error vs. epoch
- Decision boundary in 2D (project to 2 principal components)
- Training and validation error vs. epoch



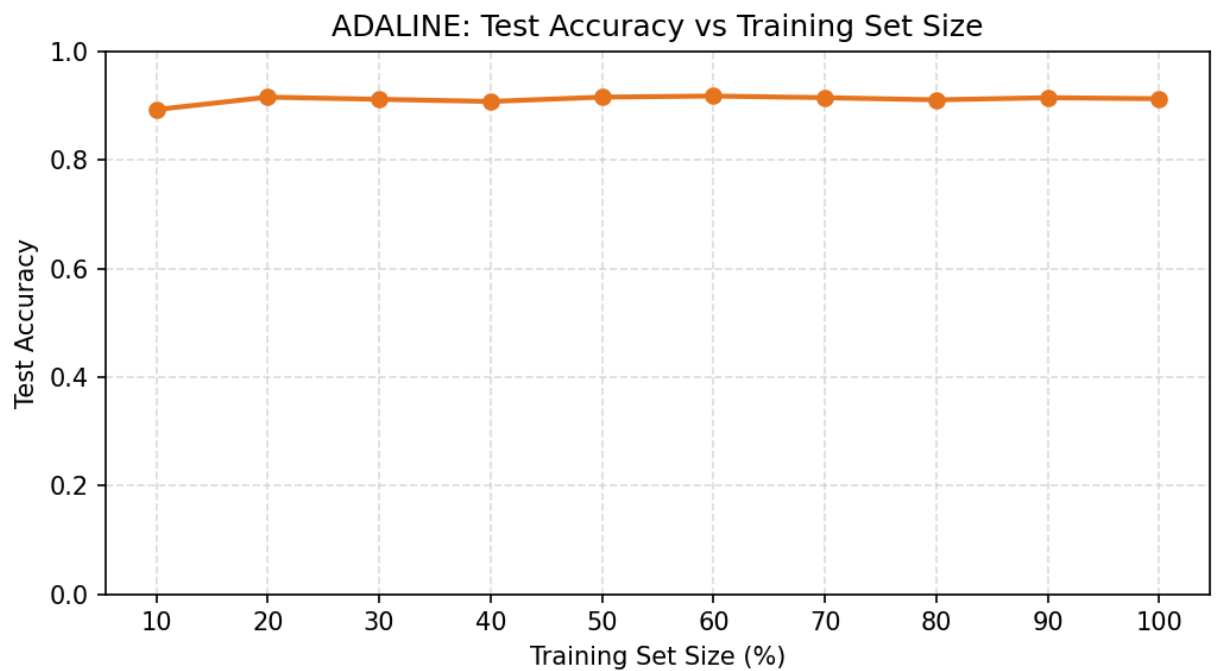
- **MSE vs. Epochs:** Training starts with a big drop in error during the first 20 epochs (from 1.0 down to about 0.37). After that, the improvements get much smaller as the curve levels off. This is classic gradient descent: it fixed the big mistakes fast, then spent the rest of the time fine-tuning.
- **Train vs. Validation:** Both the training and validation curves look almost identical and end up at basically the same value. Since there's no real gap between them, we know the model isn't overfitting—it's actually learning the patterns rather than just memorizing the data.
- **Decision Boundary:** The black line in the PCA plot cuts diagonally through the data to separate the two groups. Because the classes overlap so much in the middle, it's impossible to get a "perfect" split, but the line sits exactly where it should to capture the main difference between them.
- **The Result:** Even with that messy overlap in the center, the model still manages to hit about 91% accuracy on the test set, which is a solid result for a linear classifier.

C. Experiment with learning rates $\eta \in \{0.01, 0.1, 1.0, 10.0\}$. Plot convergence curves and discuss the effect.



1. $\eta = 0.01$ (slow convergence): The model converges to the same MSE minimum as $\eta = 0.1$, but requires the full 300 epochs to do so. The curve decays smoothly but slowly. This is a safe choice but computationally inefficient.
2. $\eta = 0.1$ (optimal): Rapid convergence within ≈ 20 –30 epochs, reaching the MSE plateau cleanly without oscillation. Both training and validation losses align closely. This is the best practical choice for this dataset.
3. $\eta = 1.0$ (diverged): The MSE explodes and the model does not converge. With large learning rates, weight updates overshoot the minimum, causing the loss to oscillate or grow without bound. Plotted as a dotted line at the upper boundary.
4. $\eta = 10.0$ (diverged): Divergence is even more immediate and severe. The step size is orders of magnitude too large for this feature scale, making learning impossible.

D. Vary the amount of training data from 10% to 100% in increments of 10%. Plot final accuracy vs. training set size.



Observations. Test accuracy is remarkably stable across training set sizes, ranging from approximately 89% at 10% of the data to 91–92% at larger fractions. Key findings:

- Even with only 10% of training data, the model achieves $\approx 89\%$ accuracy, suggesting that ADALINE's linear decision boundary can be estimated with relatively few samples.
- There is a modest improvement from 10% to 20% (approximately +2%), after which accuracy plateaus and remains essentially flat.
- No degradation occurs at larger training sizes, confirming the absence of overfitting.
- The plateau behaviour suggests the model has reached its capacity limit — the irreducible error is dominated by the inherent class overlap in the data rather than by insufficient training samples.